



Development and validation of a staging system for HPV-related oropharyngeal cancer by the International Collaboration on Oropharyngeal cancer Network for Staging (ICON-S): a multicentre cohort study

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Summary

Background Human papillomavirus-related (HPV+) oropharyngeal cancer is a rapidly emerging disease with generally good prognosis. Many prognostic algorithms for oropharyngeal cancer incorporate HPV status as a stratification factor, rather than recognising the uniqueness of HPV+ disease. The International Collaboration on Oropharyngeal cancer Network for Staging (ICON-S) aimed to develop a TNM classification specific to HPV+ oropharyngeal cancer.

Methods The ICON-S study included patients with non-metastatic oropharyngeal cancer from seven cancer centres located across Europe and North America; one centre comprised the training cohort and six formed the validation cohorts. We ascertained patients' HPV status with p16 staining or in-situ hybridisation. We compared overall survival at 5 years between training and validation cohorts according to 7th edition TNM classifications and HPV status. We used recursive partitioning analysis (RPA) and adjusted hazard ratio (AHR) modelling methods to derive new staging classifications for HPV+ oropharyngeal cancer. Recent hypotheses concerning the effect of lower neck lymph nodes and number of lymph nodes were also investigated in an exploratory training cohort to assess relevance within the ICON-S classification.

Findings Of 1907 patients with HPV+ oropharyngeal cancer, 661 (35%) were recruited at the training centre and 1246 (65%) were enrolled at the validation centres. 5-year overall survival was similar for 7th edition TNM stage I, II, III, and IVA (respectively; 88% [95% CI 74–100]; 82% [71–95]; 84% [79–89]; and 81% [79–83]; global $p=0.25$) but was lower for stage IVB (60% [53–68]; $p<0.0001$). 5-year overall survival did not differ among N0 (80% [95% CI 73–87]), N1–N2a (87% [83–90]), and N2b (83% [80–86]) subsets, but was significantly lower for those with N3 disease (59% [51–69]; $p<0.0001$). Stage classifications derived by RPA and AHR models were ranked according to survival performance, and AHR-New was ranked first, followed by AHR-Orig, RPA, and 7th edition TNM. AHR-New was selected as the proposed ICON-S stage classification. Because 5-year overall survival was similar for patients classed as T4a and T4b, T4 is no longer subdivided in the re-termed ICON-S T categories. Since 5-year overall survival was similar among N1, N2a, and N2b, we re-termed the 7th edition N categories as follows: ICON-S N0, no lymph nodes; ICON-S N1, ipsilateral lymph nodes; ICON-S N2, bilateral or contralateral lymph nodes; and ICON-S N3, lymph nodes larger than 6 cm. This resembles the N classification of nasopharyngeal carcinoma but without a lower neck lymph node variable. The proposed ICON-S classification is stage I (T1–T2N0–N1), stage II (T1–T2N2 or T3N0–N2), and stage III (T4 or N3). Metastatic disease (M1) is classified as ICON-S stage IV. In an exploratory training cohort ($n=702$), lower lymph node neck involvement had a significant effect on survival in ICON-S stage III but had no effect in ICON-S stage I and II and was not significant as an independent factor. Overall survival was similar for patients with fewer than five lymph nodes and those with five or more lymph nodes, within all ICON-S stages.

Interpretation Our proposed ICON-S staging system for HPV+ oropharyngeal cancer is suitable for the 8th edition of the Union for International Cancer Control/American Joint Committee on Cancer TNM classification. Future work is needed to ascertain whether T and N categories should be further refined and whether non-anatomical factors might augment the full classification.

Funding None.

Introduction

Human papillomavirus-related (HPV+) oropharyngeal cancer is a rapidly emerging disease in many countries that differs from tobacco-related and alcohol-related (HPV–) oropharyngeal cancer. Both diseases share similar

histological features, arise from the oropharynx, and use the 7th edition Union for International Cancer Control/American Joint Committee on Cancer (UICC/AJCC) TNM staging system, but patients with HPV+ oropharyngeal cancer have remarkably higher overall

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See [Comment](#) page 403

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Research in context

Evidence before this study

Human papillomavirus-related (HPV+) oropharyngeal cancer is a rapidly emerging disease with generally good prognosis. To date, most prognostic algorithms for oropharyngeal cancer incorporate HPV status as a stratification factor, rather than recognising the uniqueness of HPV+ disease, which affects a disparate population of patients and is anticipated to require different treatments. Opinions have converged on the principle that a novel staging system is needed urgently to properly depict the character and prognosis of HPV+ disease, which contrasts with smoking-related (ie, HPV-unrelated [HPV-]) oropharyngeal cancer, from which the 7th edition of the Union for International Cancer Control/American Joint Committee on Cancer (UICC/AJCC) TNM classification was derived. A challenge in developing a new TNM classification is to obtain an adequate study sample covering the full disease spectrum (including early to advanced stages). Most trials in head and neck cancer have, in general, addressed locally advanced disease only, and trials specific to oropharyngeal cancer are scarce. Trials also lack HPV-specific data, as do administrative and registry data sources.

Added value of this study

The International Collaboration on Oropharyngeal cancer Network for Staging (ICON-S) study represents an

international effort to develop a pretreatment TNM clinical staging classification for HPV+ oropharyngeal cancer.

The ICON-S study included almost 2000 patients with HPV+ oropharyngeal cancer from seven institutions across Europe and North America. Heterogeneity tests showed that hazard ratios for stage III versus II disease, and for stage II versus I disease, were in the same direction (>1.0 for higher stages) across all institutions, which supports applicability of the ICON-S classification across various patient populations.

Implications of all the available evidence

At the present time, most patients with HPV+ oropharyngeal cancer are told they have stage IV disease, but the reality is that their outlook is similar to that of patients with the most curable malignant diseases. This situation is alarming for many patients when first faced with their diagnosis and might perpetuate the idea that traditional intensified treatments are always needed—a notion that is being challenged. The proposed ICON-S classification for HPV+ oropharyngeal cancer permits a more appropriate depiction of prognosis of HPV+ disease than is available with the 7th edition of the UICC/AJCC TNM classification. Moreover, the ICON-S classification will enhance stratification into more appropriate groups to both facilitate translational research and optimise clinical trial design and outcome reporting.

survival than do those with HPV- disease.^{1,2} In fact, most patients with HPV+ oropharyngeal cancer at a numerically advanced stage do not experience the same poor outcome as do those with HPV- disease. For example, fewer than 50% of patients with stage IV HPV- oropharyngeal cancer will be alive at 5 years compared with more than 70% with HPV+ disease.¹ This disparity in overall survival affects clinical trial design and outcomes research and interferes with clinical decision making, because prognosis is not reflected by the 7th edition TNM staging system.^{2,3}

A novel clinical staging system based on a single-centre dataset has been proposed for HPV+ oropharyngeal cancer.¹ Favourable outcomes led to all disease classified as 7th edition TNM stage III and most disease categorised as stage IVA (T1–T3N0–N2c) to be amended to earlier disease stages. Whether this classification is generalisable across different geographic jurisdictions needs validation. Therefore, we aimed to do a multicentre study (International Collaboration on Oropharyngeal cancer Network for Staging [ICON-S]), expanding on data from the initial single centre,¹ to construct a validated clinical TNM classification for HPV+ oropharyngeal cancer applicable before any treatment—eg, surgery, radiotherapy, or chemotherapy.

Methods

Patients

We did the ICON-S study in an expanded population from the initial single centre (Princess Margaret Cancer Centre, University of Toronto),¹ with additional cohorts

from four North American cancer centres (M D Anderson Cancer Center, Memorial Sloan Kettering Cancer Center, Cleveland Clinic Taussig Cancer Institute, and University of Colorado Cancer Center) and two European centres (Rigshospitalet, Copenhagen University Hospital, and Maastricht University Medical Centre). We chose these centres based on known availability of suitable data—eg, publications and personal communications from other ongoing collaborations—and willingness to participate. We obtained ethics approval from every participating institution.

We included in our dataset all patients with newly diagnosed non-metastatic (M0) oropharyngeal cancer undergoing either primary surgery or primary radiotherapy with or without chemotherapy. We based initial staging on the 7th edition TNM staging system.^{4,5} We ascertained HPV status by either p16 staining or in-situ hybridisation, according to institutional practice. We excluded patients in whom we could not establish HPV status. We ascertained HPV status in at least 50% of patients treated up to 2011 at every participating institution.

Because the institution from the initial discovery study¹ (Princess Margaret Cancer Centre) provided the largest sample and was of sufficient size with quality-assured, prospectively gathered, clinical data,⁶ and because the original hypothesis for the ICON-S study was generated at this institution, we selected it as the training cohort. We pooled data for the six other institutions to form the validation cohort. To investigate

emerging proposals,^{7,8} we expanded the training cohort into an exploratory training cohort when detailed imaging for contemporary radiotherapy targeting (intensity-modulated radiotherapy) became available after the initial study¹ was completed.

Procedures

Treatment and follow-up surveillance took place in multidisciplinary settings and followed accepted guidelines. Generally, patients with early-stage disease received single-modality treatment, with either surgery or radiotherapy alone, whereas those in advanced stages were treated with combined modalities, including chemotherapy (induction, concurrent, or both) with radiotherapy, surgery and postoperative radiotherapy or chemotherapy, or radiotherapy combined with EGFR inhibitors (eg, cetuximab). We gathered clinical information (including smoking pack-years and outcomes) prospectively for the training cohort and for one validation centre (Denmark) and retrospectively from clinical reports for the remaining validation centres. We affirmed vital status through links with the Ontario population-based cancer registry (for the training cohort) and national registries (for data from the Netherlands and Denmark). For the four US validation centres, we confirmed vital status with institutional cancer registries or through the social security death index, or both of these.

We did investigations in an exploratory training cohort with respect to the potential effect of lower neck lymph nodes⁷ and number of lymph nodes,⁸ to assess prognostication in overall survival and recurrence-free survival and for potential refinement within the ICON-S classification. A head and neck neuroradiologist (EY) recorded the number and location of involved lymph nodes using staging CT or MRI, and a second neuroradiologist (E Bartlett, Princess Margaret Cancer Centre) audited the findings for inter-rater variability. We defined lower neck lymph node involvement as level IV or Vb (ie, extension caudal to the cricoid cartilage), as adopted for nasopharyngeal cancer⁹ and studied in oropharyngeal cancer.⁷

Statistical analysis

We compared training and validation cohorts with the Kruskal-Wallis test for continuous variables and Fisher's exact test for categorical variables. We calculated overall survival (including death from any cause) with the Kaplan-Meier method, from the date of diagnosis, and we compared data with the log-rank test. We calculated hazard ratios for risk of death (HRs) for training and validation cohorts separately by multivariable Cox regression analyses. We used two-tailed tests and judged *p* values less than 0.05 significant.

We compared overall survival by HPV status and by 7th edition TNM stage for patients with HPV+ and HPV- disease separately. We derived potential novel

stage classifications for patients with HPV+ oropharyngeal cancer from two statistical models, as described elsewhere.¹ Briefly, we used recursive partitioning analysis (RPA) to derive RPA stage, for which we judged T1, T2, T3, and T4 and N0, N1, N2a, N2b, N2c, and N3 ordinal variables; and we used adjusted hazard ratios (AHRs) to derive AHR stage, for which we calculated HRs for risk of death for every T and N combination, adjusting for age, smoking, and chemotherapy. The RPA model is based on the optimised binary partition of T or N categories, using splitting, pruning, and final tree structure selection algorithms. We developed the AHR model based on the multivariable Cox model. The AHR model calculates HRs for risk of death, adjusted for age, smoking, and treatment, with various combinations of T and N categories, taking into account minimum hazard difference, the ordinal order of T and N categories, and the sample size balance between the stage subgroups. We chose the best performing classification with respect to survival based on five established criteria.¹⁰ First, we used hazard consistency to assess the similarity of overall survival for subgroups defined by T and N within each stage group. Second, we used hazard discrimination to ascertain differences in overall survival across stage groups, to assess how equally they were spaced. Third, we used explained variance to calculate the percentage of variation in overall survival accounted for by stage groupings. Fourth, we used sample size balance to investigate the difference in sample sizes across stage groups. Finally, we used likelihood difference to assess the difference in goodness-of-fit between models. We normalised the actual score for each criterion then added the component scores to achieve a summary score. We ranked the stage grouping schemas within the summary scores, with the lowest score ranking first. After we established the new stage system, we applied meta-analysis using generic inverse-variance methods and forest plots to measure heterogeneity of the stage performance across different institutions. If heterogeneity tests were not significant, we used fixed-effect models; otherwise, we used random-effect models.

Role of the funding source

No funding was received for this study. The corresponding author had full access to all data in the study and had final responsibility for the decision to submit for publication.

Results

Between 1998–2007 and 2011, 3998 consecutive patients with oropharyngeal cancer were identified at the participating centres (recruitment start dates differed by centre and are shown in the appendix p 1). HPV status could be ascertained in 2603 (65%) patients, including 1907 with HPV+ disease and 696 with

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HPV- disease (appendix p 1). Patients' characteristics are shown in table 1. Primary radiotherapy with or without chemotherapy was used in 2553 (98%) patients, with the remainder receiving primary surgery with or without postoperative radiotherapy or chemotherapy (appendix p 2). HPV+ patients had significantly better overall survival than did those with HPV- disease (appendix p 3).

HPV- patients (n=696) had a monotonic (consistent) reduction in overall survival according to the 7th edition TNM staging system (figure 1). The proportion of patients alive at 5 years with a TNM stage of I, II, III, IVA, and IVB was 76% (95% CI 61–95), 68% (56–81), 53% (44–64), 45% (40–50), and 34% (25–48), respectively. By contrast, there was little difference in 5-year overall

survival for HPV+ patients (n=1907) with stage I, II, III, and IVA disease (global p=0·25), but it was significantly lower for those with stage IVB disease (p<0·0001). The proportion of HPV+ patients alive at 5 years with stage I, II, III, IVA, and IVB disease was 88% (95% CI 74–100), 82% (71–95), 84% (79–89), 81% (79–83), and 60% (53–68), respectively. Overall survival of patients with HPV+ oropharyngeal cancer was similar between T4a (n=231) and T4b (n=44) subsets (p=0·41) and was significantly lower than that of the T1–T3 subset (vs T4, p<0·0001). The proportion of patients alive with HPV+ oropharyngeal cancer at 5 years with T1, T2, T3, T4a, and T4b disease was 89% (95% CI 87–92), 83% (80–87), 76% (72–81), 58% (51–65), and 57% (44–75), respectively. Patients with HPV+ N3 disease had significantly lower

	HPV+				HPV-			
	Total (n=1907)	Training cohort (n=661)	Validation cohort (n=1246)	p	Total (n=696)	Training cohort (n=291)	Validation cohort (n=405)	p
Follow-up (years)	4·9 (3·1–6·0)	5·5 (3·2–6·6)	4·6 (3·1–5·5)	..	5·2 (1·2–5·5)	4·6 (1·0–5·0)	5·5 (1·4–5·8)	..
Sex	..	..	..	0·0009	..	..	..	0·150
Female	314 (16%)	135 (20%)	179 (14%)	..	169 (24%)	79 (27%)	90 (22%)	..
Male	1593 (84%)	526 (80%)	1067 (86%)	..	526 (76%)	212 (73%)	315 (78%)	..
Age (years)	57 (51–63)	57 (51–65)	56 (51–62)	0·0001	61 (55–68)	65 (58–72)	59 (53–65)	<0·0001
Smoking pack-years	8 (0–30)	15 (0–30)	6 (0–25)	0·0031	35 (20–50)	40 (25–50)	30 (13–50)	0·0006
Unknown	18	0	18	..	5	1	4	..
T category	..	..	..	<0·0001	..	..	..	<0·0001
T1	504 (26%)	135 (20%)	369 (30%)	..	107 (15%)	30 (10%)	77 (19%)	..
T2	716 (38%)	234 (35%)	482 (39%)	..	201 (29%)	99 (34%)	102 (25%)	..
T3	412 (22%)	175 (26%)	237 (19%)	..	183 (26%)	78 (27%)	105 (26%)	..
T4a	231 (12%)	87 (13%)	144 (11%)	..	176 (25%)	60 (21%)	117 (29%)	..
T4b	44 (2%)	30 (5%)	14 (1%)	..	28 (4%)	24 (8%)	4 (1%)	..
N category	..	..	..	<0·0001	..	..	..	0·0013
N0	173 (9%)	81 (12%)	92 (7%)	..	153 (22%)	88 (30%)	65 (16%)	..
N1	216 (11%)	66 (10%)	150 (12%)	..	104 (15%)	41 (14%)	63 (16%)	..
N2a	200 (10%)	50 (8%)	150 (12%)	..	35 (5%)	9 (3%)	26 (6%)	..
N2b	749 (39%)	241 (36%)	508 (41%)	..	192 (28%)	65 (22%)	128 (32%)	..
N2c	436 (23%)	167 (25%)	269 (22%)	..	172 (25%)	70 (24%)	102 (25%)	..
N3	133 (7%)	56 (8%)	77 (6%)	..	39 (6%)	18 (6%)	21 (5%)	..
7th edition TNM stage	..	..	..	0·0004	..	..	..	<0·0001
I	19 (1%)	9 (1%)	10 (1%)	..	27 (4%)	13 (4%)	14 (3%)	..
II	71 (4%)	29 (4%)	42 (3%)	..	62 (9%)	40 (14%)	22 (5%)	..
III	253 (13%)	88 (13%)	165 (13%)	..	107 (15%)	46 (16%)	61 (15%)	..
IVA	1392 (73%)	451 (68%)	941 (76%)	..	433 (62%)	151 (52%)	283 (70%)	..
IVB	172 (9%)	84 (13%)	88 (7%)	..	66 (9%)	41 (14%)	25 (6%)	..
Primary treatment	..	..	..	0·0033	..	..	..	0·61
Surgery	34 (2%)	4 (1%)	30 (2%)	..	16 (2%)	8 (3%)	8 (2%)	..
Radiotherapy	1873 (98%)	657 (99%)	1216 (98%)	..	680 (98%)	283 (97%)	397 (98%)	..
Chemotherapy	..	..	..	<0·0001	..	..	..	<0·0001
Yes	1260 (66%)	321 (49%)	939 (75%)	..	316 (45%)	80 (27%)	237 (58%)	..
No	647 (34%)	340 (51%)	307 (25%)	..	379 (55%)	211 (73%)	168 (42%)	..

Data are median (IQR) or number of patients (%). Differences in demographic and clinical characteristics between training and validation cohorts were compared with the Kruskal-Wallis test for continuous variables and Fisher's exact test for categorical variables.

Table 1: Patients' characteristics

overall survival compared with those with HPV+ N0–N2c disease ($p<0.0001$). The proportion of HPV+ patients alive at 5 years with N0, N1–N2a, N2b, N2c, and N3 disease was 80% (95% CI 73–87), 87% (83–90), 83% (80–86), 74% (70–79), and 59% (51–69), respectively (figure 1). The proportion of HPV– patients alive at

5 years with T1, T2, T3, T4a, and T4b disease was 71% (95% CI 62–80), 57% (51–65), 39% (31–47), 34% (27–42), and 45% (30–69), respectively, and with N0, N1–N2a, N2b, N2c, and N3 disease was 54% (47–63), 58% (50–67), 51% (44–59), 37% (30–46), and 28% (17–47), respectively.

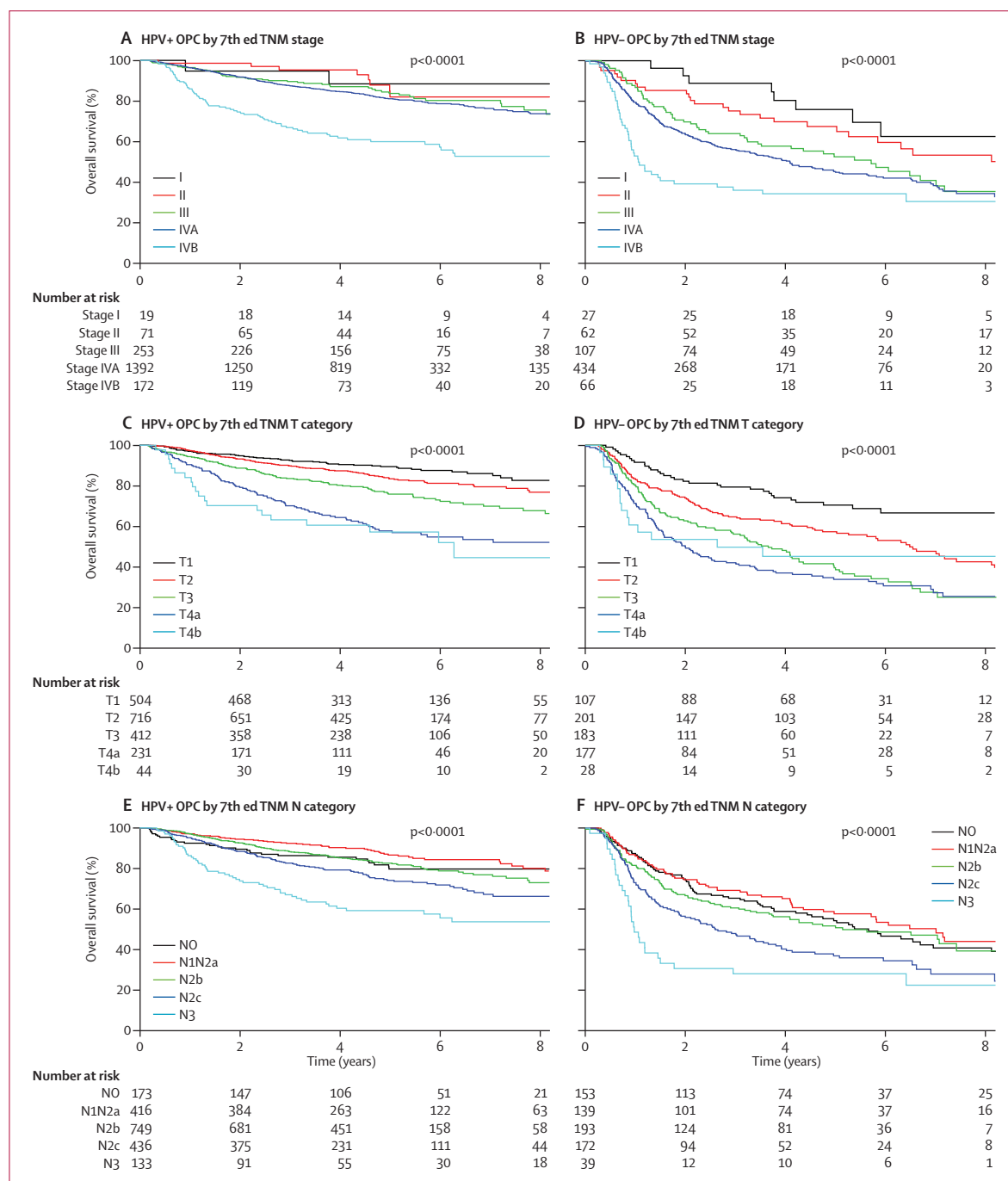


Figure 1: Kaplan-Meier estimates of overall survival by (A, B) 7th edition TNM stage, (C, D) T categories, and (E, F) N categories, for the entire cohort Global p values are the test for overall trend on multiple categories of the predictor in the regression model. Hazard ratios (HRs) were adjusted for age, smoking pack-years, and use of systemic agents (eg, EGFR inhibitors, cytotoxic chemotherapy). HPV+=human papillomavirus-related. HPV-=human papillomavirus-unrelated. OPC=oropharyngeal cancer.

	HPV+			HPV-		
	Hazard ratio (95% CI)	p*	Global p†	Hazard ratio (95% CI)	p*	Global p†
7th edition TNM stage	..	..	<0.0001	..	..	<0.0001
I	Reference	..	..	Reference	..	..
II	1.19 (0.25–5.62)	0.83	..	1.62 (0.74–3.54)	0.23	..
III	2.52 (0.61–10.41)	0.2	..	2.91 (1.38–6.13)	0.005	..
IVA	3.41 (0.84–13.85)	0.086	..	4.80 (2.34–9.82)	<0.0001	..
IVB	7.91 (1.92–32.59)	0.004	..	6.34 (2.96–13.6)	<0.0001	..
Age	1.03 (1.02–1.04)	..	<0.0001	1.01 (1.00–1.03)	..	0.014
Smoking pack-year	1.01 (1.01–1.02)	..	<0.0001	1.01 (1.00–1.01)	..	0.0016
Systemic agent	..	..	<0.0001	..	..	<0.0001
None	Reference	..	..	Reference	..	..
EGFR inhibitor	0.73 (0.49–1.08)	0.12	..	0.44 (0.26–0.75)	0.003	..
Chemotherapy	0.52 (0.42–0.65)	<0.0001	..	0.47 (0.37–0.59)	<0.0001	..

*Pairwise comparison with reference category, based on Wald test in the Cox model. †Test for overall trend on multiple categories of the predictor in the regression model.

Table 2: Multivariable analysis of risk of death

	T1	T2	T3	T4
Training cohort				
N0	1.00, n=9	1.69 (0.20–14.53), n=29	4.24 (0.54–33.16), n=29	6.01 (0.73–49.14), n=14
N1	1.96 (0.20–17.50), n=21	3.45 (0.44–27.24), n=26	5.60 (0.67–46.72), n=12	10.01 (1.03–97.25), n=7
N2a	0.80 (0.05–12.80), n=17	0.73 (0.05–11.68), n=22	3.22 (0.29–35.71), n=8	5.09 (0.32–82.02), n=3
N2b	2.27 (0.28–18.55), n=54	2.69 (0.36–20.21), n=99	2.46 (0.31–19.52), n=59	9.55 (1.25–73.17), n=29
N2c	4.07 (0.47–34.92), n=19	4.03 (0.5–32.11), n=45	4.57 (0.59–35.14), n=49	9.57 (1.29–71.21), n=54
N3	8.90 (1.08–73.08), n=15	6.56 (0.76–56.85), n=13	17.44 (2.24–135.71), n=18	5.97 (0.70–50.99), n=10
Validation cohort				
N0	1.00, n=10	0.85 (0.09–8.26), n=42	2.21 (0.23–21.49), n=25	2.27 (0.2–25.68), n=15
N1	0.53 (0.05–5.86), n=54	2.20 (0.27–18.05), n=58	3.03 (0.35–26.54), n=28	5.16 (0.52–50.81), n=10
N2a	1.09 (0.13–9.14), n=87	1.71 (0.19–15.47), n=43	4.64 (0.47–46.13), n=15	23.12 (2.04–262.3), n=5
N2b	2.11 (0.28–16.02), n=161	3.55 (0.48–26.34), n=230	4.53 (0.58–35.12), n=79	5.71 (0.72–45.19), n=38
N2c	1.00 (0.09–11.26), n=42	3.37 (0.43–26.65), n=84	5.56 (0.72–42.71), n=70	8.63 (1.13–66.01), n=73
N3	9.31 (1.06–81.53), n=15	3.32 (0.33–32.95), n=25	9.56 (1.12–81.55), n=20	23.60 (2.88–193.46), n=17

Data are hazard ratio (95% CI), number of patients. Hazard ratios were adjusted for age, smoking pack-year, and use of cytotoxic chemotherapy (yes vs no).

Table 3: Hazard ratios for risk of death by T–N combination

Multivariable Cox regression analyses showed poor prognostic discrimination of 7th edition TNM stages I, II, III, and IVA in patients with HPV+ oropharyngeal cancer (table 2). Moreover, EGFR inhibitors had no survival effect compared with radiotherapy alone or surgery with or without postoperative radiotherapy in patients with HPV+ disease, whereas these agents reduced mortality risks in individuals with HPV– oropharyngeal cancer. Chemotherapy and smoking pack-years affected overall survival in patients with both HPV+ and HPV– oropharyngeal cancer (table 2).

Novel staging schemas constructed with previously established RPA and AHR models¹ were assessed for HPV+ patients in the training cohort and compared with those in the validation cohort. Because the proportions of patients alive at 5 years in the T4a and T4b categories (7th edition TNM) were almost identical in

both the training (51% [95% CI 41–64] vs 55% [40–77]) and validation (62% [53–71] vs 61% [39–95]) cohorts, they were grouped as one T4 category. Furthermore, EGFR inhibitors were included in the no chemotherapy group (radiotherapy alone or surgery with or without postoperative radiotherapy) for comparison with the cytotoxic chemotherapy group (concurrent chemotherapy or postoperative chemotherapy). Table 3 presents HRs for risk of death by 7th edition T–N combination, adjusted for age, smoking pack-years, and use of chemotherapy. Patients with N0 disease (7th edition) seemed to fare slightly worse with respect to risk of death than did those with N2a disease, but this was not significant after adjustment for clinical confounders such as T category and treatment (training cohort, N2a vs N0: HR 0.63, 95% CI 0.23–1.77, p=0.38; validation cohort, N2a vs N0: 1.92, 0.70–5.31, p=0.21; table 3).

RPA stage classification	T1	T2	T3	T4
N0	I	I	I	III
N1	I	I	I	III
N2a	I	I	I	III
N2b	I	I	I	III
N2c	II	II	II	III
N3	III	III	III	III

AHR-Orig stage classification	T1	T2	T3	T4
N0	I	I	II	III
N1	I	I	II	III
N2a	I	I	II	III
N2b	I	II	II	IVA
N2c	II	II	III	IVA
N3	III	III	IVA	IVA

AHR-New stage classification	T1	T2	T3	T4
N0	I	I	II	III
N1	I	I	II	III
N2a	I	I	II	III
N2b	I	I	II	III
N2c	II	II	II	III
N3	III	III	III	III

Figure 2: Novel stage tabulation grids

Top panel: RPA stage classification. Middle panel: AHR-Orig stage classification. Lower panel: AHR-New stage classification. RPA=model derived by recursive partitioning analysis. AHR-Orig=staging schema derived with adjusted hazard ratios in initial study.¹ AHR-New=new staging schema derived with adjusted hazard ratios.

The staging schema derived with the RPA model (figure 2) was named RPA stage I (corresponding to T1–T3N0–N2b), RPA stage II (T1–T3N2c), and RPA stage III (T4 or N3). The proportion of patients alive at 5 years with RPA stage I was 83% (95% CI 79–87) in the training cohort and 87% (84–89) in the validation cohort; respective proportions for RPA stage II were 79% (72–87) and 82% (76–89), and for RPA stage III were 53% (46–62) and 65% (58–72; figure 3).

The AHR model derived two stage schemas: one was the same as in the previous discovery study,¹ termed AHR-Original (AHR-Orig); and the other was called AHR-New (figure 2). With AHR-Orig, the HPV+ cohort was subdivided into AHR-Orig stage I (corresponding to T1N0–N2b or T2N0–N2a), AHR-Orig stage II (T1N2c, T2N2b–N2c, or T3N0–N2b), AHR-Orig stage III (T1–T2N3, T3N2c, or T4N0–N2a), and AHR-Orig stage IVA (T3N3 or T4N2b–N3). The proportion of patients alive at 5 years in the training cohort was 85% (95% CI 79–90) for AHR-Orig stage I, 81% (76–86) for AHR-Orig stage II, 68% (59–78) for AHR-Orig stage III, and 51% (42–62) for AHR-Orig stage IVA (figure 3). Respective proportions in the validation cohort were 91% (88–94), 83% (79–87), 73% (65–82), and 60% (52–70). With AHR-New, the HPV+ cohort was subdivided into AHR-New stage I (corresponding to T1–T2N0–N2b),

AHR-New stage II (T1–T2N2c or T3N0–N2c), and AHR-New stage III (T4 or N3). The proportion of patients alive at 5 years in the training cohort was 85% (95% CI 81–90) for AHR-New stage I, 78% (73–84) for AHR-New stage II, and 53% (46–62) for AHR-New stage III (figure 3). Respective proportions in the validation cohort were 88% (85–91), 81% (77–86), and 65% (58–72). Table 4 presents HRs for risk of death for the three staging classifications, adjusted for age, smoking pack-years, and use of chemotherapy.

The appendix (p 4) shows how the RPA, AHR-Orig, and AHR-New staging classifications performed with respect to survival compared with the 7th edition TNM staging system. AHR-New performed best in both training and validation cohorts; AHR-Orig had good performance in the training cohort but ranked slightly lower than AHR-New in the validation cohort. RPA was ranked consistently in third place for both training and validation cohorts, and the 7th edition TNM staging system was least useful.

In view of the good performance in both training and validation cohorts of the AHR-New staging system (appendix p 4), and its practical TNM stage tabulation grid (figure 2), the AHR-New staging system was selected as the ICON-S recommended system (herein referred to as ICON-S stage) and represents the proposed 8th edition TNM. Meta-analysis of ICON-S stage II versus stage I, and ICON-S stage III versus stage II, showed good coherence (HR all >1.0) across all participating centres, except for the centre with the smallest sample size (appendix p 5).

Table 5 shows the re-termed ICON-S T and N categories and stage classification. Because the proportions of patients alive at 5 years with N1–N2a and N2b disease (ie, unilateral neck) were almost identical (87% [95% CI 83–90] vs 83% [80–86]), they were merged into one N category (ICON-S N1). N2c disease (bilateral or contralateral neck) was classified as ICON-S N2, and lymph nodes larger than 6 cm remain as ICON-S N3. Further, because overall survival was almost identical for patients with T4a and T4b disease (58% [95% CI 51–65] vs 57% [44–75]), T4 is no longer subdivided in the re-termed ICON-S T categories (table 5). The ICON-S stage tabulation grid, which represents the proposed 8th edition TNM, is summarised in figure 4.

Based on ICON-S N categories, the entire cohort of 1907 patients with HPV+ oropharyngeal cancer were divided into ICON-S stage I (n=962; T1–T2N0–N1), ICON-S stage II (n=564; T1–T2N2 or T3N0–N2), and ICON-S stage III (n=381; T4 or N3). Distant metastatic disease (M1) was regarded as the most advanced stage a priori,¹ because of its known adverse prognosis irrespective of T and N category,¹¹ and can be classified as ICON-S stage IV. HRs for ICON-S N1, N2, and N3 showed good coherence with declining outcome as ICON-S N categories increased within each T category

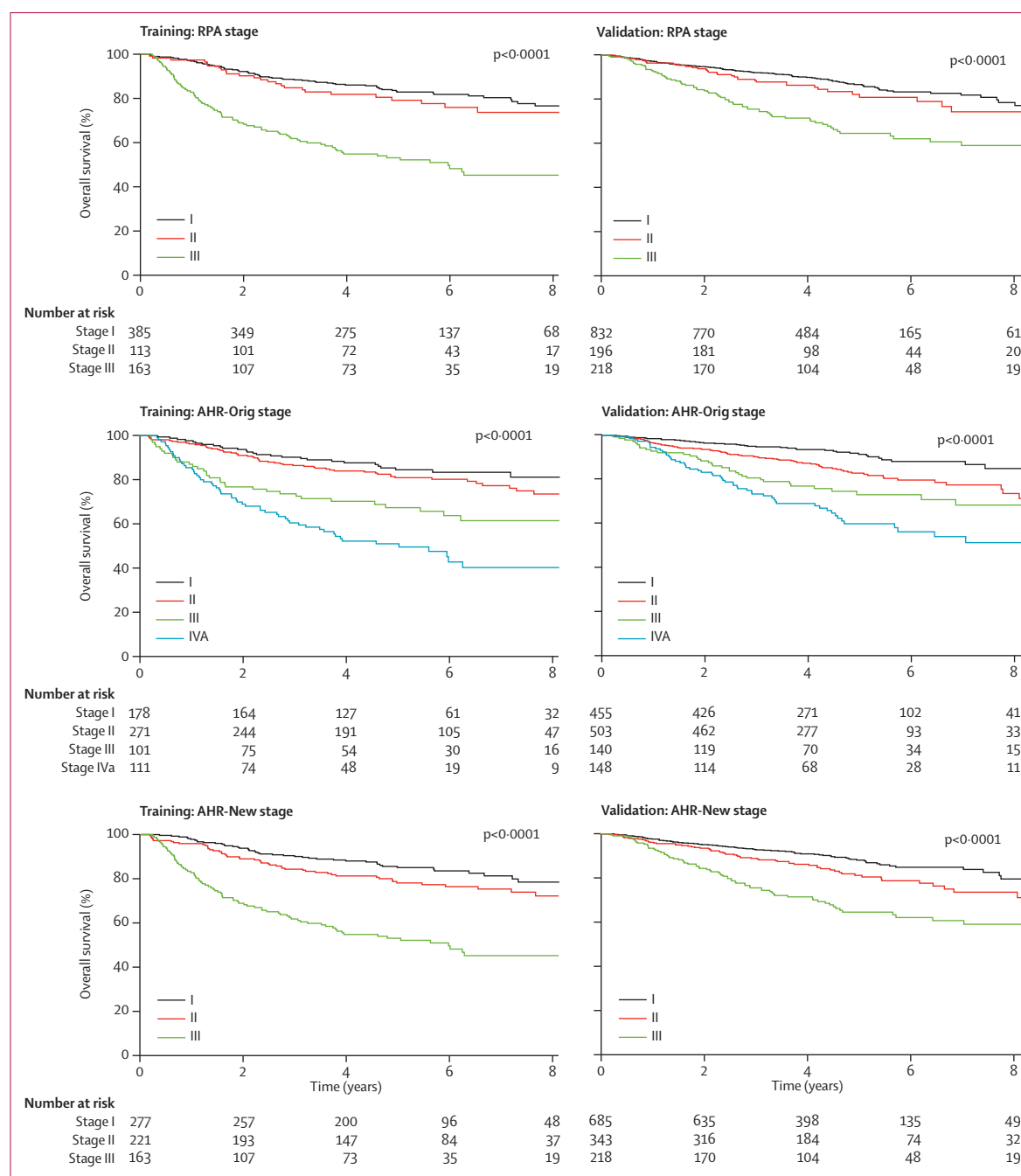


Figure 3: Kaplan-Meier estimates of overall survival in the training and validation cohorts, by stage classification

Global p values are presented to compare overall survival across stages. RPA=model derived by recursive partitioning analysis. AHR-Orig=staging schema derived with adjusted hazard ratios in initial study; AHR-New=new staging schema derived with adjusted hazard ratios.

(table 5), except for a non-significantly higher HR for ICON-S T2N1 versus T1N2 (HR 1.30, 95% CI 0.60–2.82; $p=0.51$).

Between 2005 and 2013, 1122 patients were recruited to the exploratory training cohort, of whom 975 (87%) were tested for HPV and 702 were HPV+. Interobserver variation suggested good concordance in number and

location of lymph nodes between both neuroradiologists ($\kappa=0.86$). Multivariable Cox regression analyses showed that presence of lower neck lymph nodes reduced overall survival (HR 1.82, 95% CI 1.22–2.70; $p=0.003$) and relapse-free survival (1.75, 1.17–2.60; $p=0.007$), whereas five or more involved lymph nodes did not affect overall survival (1.06, 0.71–1.57; $p=0.78$) but reduced relapse-free

	RPA stage		AHR-Orig stage		AHR-New stage	
	Hazard ratio (95% CI)	p	Hazard ratio (95% CI)	p	Hazard ratio (95% CI)	p
Training cohort						
Stage I	Reference	..	Reference	..	Reference	..
Stage II	1.66 (1.06–2.59)	0.026	1.68 (1.07–2.65)	0.024	1.74 (1.17–2.58)	0.006
Stage III	3.54 (2.53–4.93)	<0.0001	3.07 (1.85–5.1)	<0.0001	4.09 (2.81–5.96)	<0.0001
Stage IVA	NA	..	5.12 (3.2–8.2)	<0.0001	NA	..
Validation cohort						
Stage I	Reference	..	Reference	..	Reference	..
Stage II	1.42 (0.95–2.12)	0.089	2.20 (1.48–3.27)	<0.0001	1.64 (1.15–2.34)	0.006
Stage III	2.83 (2.03–3.93)	<0.0001	3.45 (2.12–5.60)	<0.0001	3.24 (2.27–4.62)	<0.0001
Stage IVA	NA	..	5.66 (3.60–8.91)	<0.0001	NA	..
Total						
Stage I	Reference	..	Reference	..	Reference	..
Stage II	1.56 (1.16–2.10)	0.0033	2.00 (1.49–2.68)	<0.0001	1.73 (1.33–2.23)	<0.0001
Stage III	3.30 (2.63–4.15)	<0.0001	3.39 (2.41–4.79)	<0.0001	3.78 (2.95–4.86)	<0.0001
Stage IVA	NA	..	5.67 (4.12–7.79)	<0.0001	NA	..

Hazard ratios were adjusted for age, smoking pack-years, and use of chemotherapy. p values are pairwise comparisons with the reference category, based on the Wald test in the Cox model. NA=not applicable. RPA=model derived by recursive partitioning analysis. AHR-Orig=staging schema derived with adjusted hazard ratios in initial study.¹ AHR-New=new staging schema derived with adjusted hazard ratios.

Table 4: Hazard ratios for risk of death for the three stage classifications

	7th edition TNM N category	ICON-S N category	ICON-ST category			
			T1	T2	T3	T4*
Gross lymph node						
None	N0	N0	1.00, n=19	1.20 (0.25–5.65), n=71	3.41 (0.77–15.16), n=54	4.33 (0.93–20.21), n=29
Unilateral neck, <6 cm	N1, N2a, N2b	N1	1.57 (0.38–6.54), n=394	2.84 (0.69–11.60), n=478	3.56 (0.85–14.83), n=201	7.30 (1.74–30.65), n=92
Bilateral or contralateral neck, <6 cm	N2c	N2	2.13 (0.44–10.31), n=61	3.51 (0.81–15.12), n=129	4.99 (1.18–20.99), n=119	9.07 (2.19–37.66), n=127
>6 cm	N3	N3	8.85 (1.97–39.85), n=30	4.88 (1.03–23.19), n=38	13.04 (3.00–56.74), n=38	11.47 (2.60–50.59), n=27

Data are hazard ratio (95% CI), number of patients. Hazard ratios are adjusted for age, smoking pack-years, and use of cytotoxic chemotherapy (yes vs no). ICON-S=International Collaboration on Oropharyngeal cancer Network for Staging. *ICON-ST definitions are unchanged from the 7th edition TNM classification, except there is no subdivision within T4 because survival was identical between T4a and T4b.

Table 5: Re-termed ICON-ST and N categories and corresponding hazard ratios for risk of death within each T category

ICON-S stage classification	T1	T2	T3	T4
N0	I	I	II	III
N1	I	I	II	III
N2	II	II	II	III
N3	III	III	III	III

Figure 4: Proposed ICON-S stage tabulation grid for 8th edition TNM
Note that distant metastatic disease (M1) is considered stage IV.

survival (1.67, 1.11–2.51; $p=0.013$), after adjusting for age, smoking, chemotherapy, and ICON-S stage (appendix pp 6, 7). However, the effect of the presence of lower neck lymph nodes was only significant within ICON-S stage III (ie, T4 or N3; HR 2.4, 95% CI 1.37–4.21; $p=0.002$) and was non-significant for ICON-S stage I ($p=0.83$) and II ($p=0.29$). The proportion of patients with ICON-S stage III disease alive at 5 years with and without lower neck lymph

nodes was 39% (95% CI 26–58) and 67% (58–77), respectively ($p=0.011$), and proportions without relapse were 49% (32–62) and 77% (67–84), respectively ($p=0.0003$). The proportion of patients alive at 5 years with five or more lymph nodes was 55% (44–69), and for those with fewer than five lymph nodes it was 60% (49–73; $p=0.94$); respective proportions without relapse were 59% (47–69) and 78% (65–86; $p=0.012$). The survival effect of lower neck lymph nodes on ICON-S stage III correlated with T4 category (41 [32%] of 128 patients with lower neck lymph nodes: the proportion alive at 5 years was 34% [95% CI 20–59] vs 66% [56–78]; $p=0.027$; in multivariable Cox regression analyses, HR 2.9, 95% CI 1.5–5.6; $p=0.0015$) and N3 category (22 [41%] of 54 patients with lower neck lymph nodes: the proportion alive at 5 years was 52% [95% CI 33–82] vs 63% [45–87]; $p=0.57$; in multivariable Cox regression analyses, HR 1.2, 95% CI 0.5–2.9; $p=0.75$; appendix pp 6, 7).

Discussion

The findings of the multicentre ICON-S have created a new, validated clinical staging system for HPV+ oropharyngeal cancer to be proposed for the 8th edition UICC/AJCC TNM classification. The current 7th edition TNM stage system is inadequate for HPV+ oropharyngeal cancer to depict prognosis, although it is acceptable for HPV- disease. Therefore, a new clinical staging system is necessary and feasible for HPV+ oropharyngeal cancer. Our results, showing similar overall survival for N1–N2b and for T4a–T4b, suggest that 7th edition TNM categories for HPV+ oropharyngeal cancer could be re-termed such that N0 corresponds with no lymph nodes, N1 with ipsilateral lymph nodes, N2 with bilateral or contralateral lymph nodes, and N3 with lymph nodes larger than 6 cm. For T categories, ICON-S terms remain the same as in the 7th edition of the TNM staging system but without subdivision of T4 into T4a and T4b. More importantly, a valid TNM classification, derived objectively from statistical models and using the re-termed ICON-S N categories, can be defined as ICON-S stage I (T1–T2N0–N1), ICON-S stage II (T1–T2N2 or T3N0–N2), and ICON-S stage III (T4 or N3). Distant metastatic disease (M1) was regarded as the most advanced stage *a priori*,¹ because of its known adverse prognosis irrespective of T and N category,¹¹ and can be classified as ICON-S stage IV.

The proposed ICON-S classification is based on 1907 patients with HPV+ oropharyngeal cancer from seven cancer centres in North America and Europe who were mostly treated non-surgically, representing the mainstay of current clinical practice.^{12–14} The derivation used a training cohort (an updated sample from 573 patients in the initial study¹ to 661 patients) and was validated in an independent, multicentre, pooled, validation cohort. In the process of designing the study, we selected the institution involved in the initial study as the training cohort to minimise potential heterogeneity arising from unforeseen outcome confounders, which might have been difficult to adjust for during stage construction. We recognise that an alternative strategy could have used random sample selection for training with potential statistical power advantages. However, the single institutional training cohort comprised a sufficiently large sample size to provide adequate power for construction of the new staging system, and for stage model performance evaluation. Heterogeneity assessment shows that the proposed ICON-S classification performs well across all seven institutions in different jurisdictions, despite demographic and treatment variability. HPV testing consisted largely of p16 staining, a well-established cost-effective surrogate for HPV status in oropharyngeal cancer compared with other methods (eg, in-situ hybridisation or PCR), if scored and interpreted appropriately.^{15–17} These principles support the generalisability and applicability of the ICON-S classification in HPV+ disease. However, our study

findings should be interpreted in the context of potential weaknesses, such as non-uniform HPV testing methods, variable proportions of HPV ascertainment across institutions, and potential biases in study populations from each institution with known versus unknown HPV status.

The proposed ICON-S stage is similar to that reported previously,¹ with the refinement that T3N0–N2b is relegated from stage I to ICON-S stage II. This change resulted from increased power with an additional year (2011), which uncovered a difference between T3N0–N2b and T1–T2N0–N2b. RPA stage in our study remained the same as in the initial study¹—ie, T1–T3N0–N2b was classified as RPA stage I. However, the performance evaluation indicated that the AHR models yielded a better staging schema compared with the RPA model. The RPA model takes into account every variable sequentially (not accounting for various T–N combinations), without adjusting for other confounding factors, whereas the AHR models account for other prognostic factors simultaneously, reflecting a more realistic outcome of differing anatomical disease extent—ie, the effect of T and N classification.

The current ICON-S stage is derived from T and N categories in the 7th edition UICC/AJCC TNM staging system. Within the ICON-S classification, T categories show a consistent survival trend across each N category level and seem not to need modification, other than elimination of the subcategories T4a and T4b. Moreover, the threshold of HR change seems to be evident at the transition from T2 to T3 in this updated cohort (meaning T3 confers higher risk than originally appreciated).¹ In essence, T3 can be separated from T1–T2 in the N0–N2b subset from ICON-S stage I. Finally, T4 is an adverse variable irrespective of N category, as is N3 irrespective of T category. Together, they represent the highest risk M0 group.

HRs for the original (7th edition TNM) N categories within each T category show inconsistency within the lower N categories (N0, N1, N2a, and N2b). This disparity manifests within T1 and T2, because of more favourable outcomes for N2a, whereas T3 seems to override it in relative terms. N2c seems homogeneous with T3 and to belong as one group (ICON-S stage II). The unexpectedly favourable effect of N2a is apparent across training and validation cohorts, and has been reported by other groups.^{18,19} The N2a effect suggests that the current N classification based on lymph node size might not adequately reflect HPV+ tumour clonogenic density in the neck. Cystic formation, a common feature in HPV+ disease,^{20,21} might dilute tumour clonogenic burden in traditional N classification. PET-CT or radiomics²² could potentially better classify tumour burden in this disease in the future. Moreover, the initial impression of better prognosis of N2a over the lower N categories disappeared when adjusted for other factors—eg, T category and age. The non-significant difference among lower N categories

additionally supports collapsing N1, N2a, and N2b into ICON-S N1, although N0 is maintained in accordance with TNM convention.

The proposed ICON-S N classification mirrors that of nasopharyngeal cancer, another virus-related cancer, except for inclusion of lower neck lymph nodes within N3 in nasopharyngeal cancer. Lower neck lymph nodes conferred increased risk of distant metastasis in oropharyngeal cancer in one report,⁷ and the similarities make it tempting to adopt the nasopharyngeal cancer N classification for HPV+ oropharyngeal cancer. Similar to nasopharyngeal cancer, analysis of the exploratory training cohort shows that lower neck lymph nodes also affect survival in HPV+ oropharyngeal cancer. However, the effect lacks independence and is typically linked immutably to more advanced disease (ICON-S stage III), which in turn drives adverse behaviour. Therefore, we are unable to provide unequivocal evidence to incorporate a lower neck lymph node variable for HPV+ oropharyngeal cancer staging at this time.

The exploratory training cohort confirmed that five or more lymph nodes identified on CT or MRI increased the risk of recurrence, as reported in a surgical series in which the number of lymph nodes was identified in neck dissection specimens.⁸ Again, the risk only exists in ICON-S stage III (7th edition TNM T4 or N3) and did not affect ICON-S stage I and II, and overall survival did not differ significantly within all ICON-S stages. Thus, incorporation of this variable into the clinical (pretreatment) aspect of the current ICON-S classification seems impractical. Inclusion of absolute number of lymph nodes would lack precedence in any clinical TNM classification for head and neck cancer, and this variable is not used traditionally for other anatomical sites, unless surgery is the primary management, to permit accurate enumeration of pathologically involved lymph nodes.

Since the main focus of the ICON-S study was to derive an anatomical stage classification, we did not include smoking pack-year as a variable to derive prognostic grouping. However, we did adjust for this variable during derivation of the ICON-S classification because of its known effect on survival. The study of non-anatomical factors, including smoking, and their incorporation into staging and prognostic algorithms is an important but still nascent domain under active research by many groups, including the Radiation Therapy Oncology Group,^{23,24} UICC,²⁵ and AJCC,²⁶ and individual institutions.^{1,27–29}

Treatment information is included in this analysis to adjust for its effect in these models to propose the ICON-S TNM classification for HPV+ oropharyngeal cancer and permit better prediction of overall survival under current treatment schedules. Ultimately, the TNM classification provides a framework for classification but does not determine treatment. It is a principle of the UICC and the AJCC that treatments remain unspecified for a staging system, and clinical TNM is an important pretreatment model applicable to any treatment approach meeting

acceptable guidelines for the disease among the medical community. Moreover, although the ICON-S classification is derived mainly from radiation-treated patients, it is also applicable to surgical patients since such cases also need clinical staging before surgery to inform this decision. We anticipate that ICON-S will contribute a better TNM classification for HPV+ oropharyngeal cancer to enhance stratification into more valid groups to facilitate and optimise clinical trials, translational studies, and outcome reporting for the future.

In conclusion, we have proposed a novel clinical stage classification (ICON-S) for HPV+ oropharyngeal cancer for the upcoming 8th edition UICC/AJCC TNM. The ICON-S stage classification predicts prognosis in different jurisdictions. Historical data can be converted readily into the ICON-S classification. We also anticipate that the ICON-S TNM will conveniently address the other aims of staging beyond prognostication, including clinical trials eligibility and stratification, monitoring adherence to clinical guidelines, assessing treatment outcome, facilitating translational research, and supporting cancer control activities.³⁰

Contributors

All authors contributed to study design, data collection, data interpretation, and writing of the report. BO'S, SHH, JS, and WX contributed to data analysis.

Declaration of interests

JF reports non-financial support from Merck Serono, outside the submitted work. CAK reports personal fees and non-financial support from Merck Serono, outside the submitted work. FH reports personal fees from Roche, outside the submitted work. All other authors declare no competing interests.

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